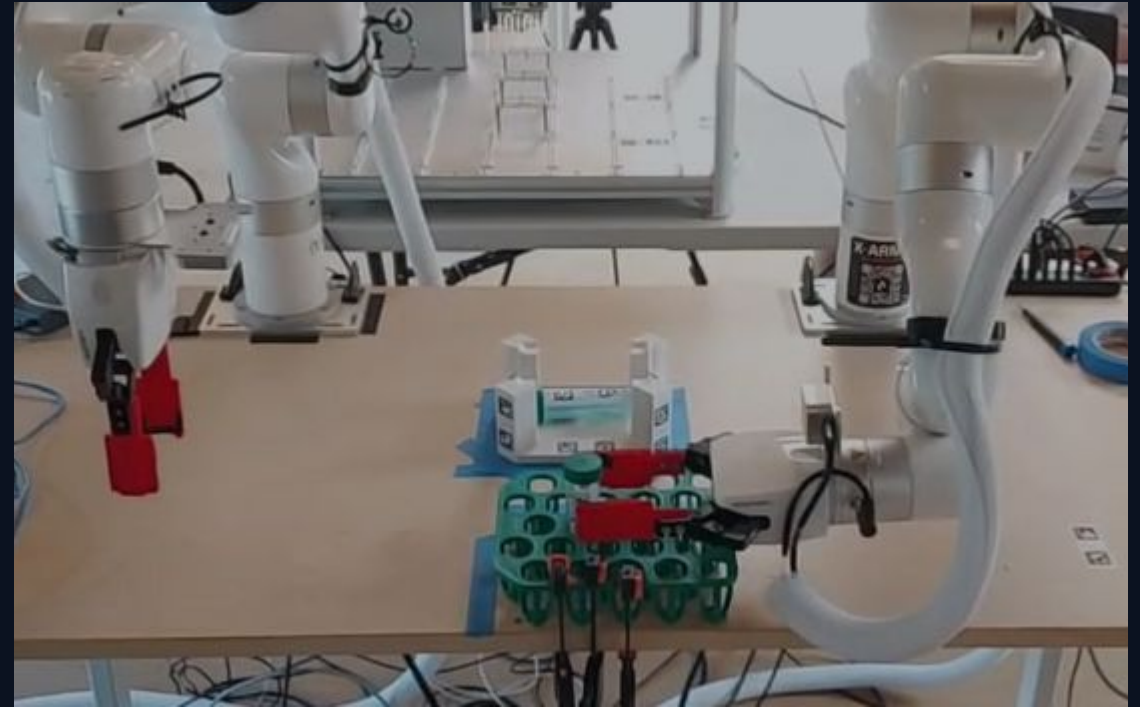


The Robust, Collaborative Lab Cell

Cooperative Dexterity + AI Physical Verification
Live on real lab hardware.



OT OneCapped
Dale Herzog · Di Hu · Lukas Bromig

Lab automation has two gaps

1

Unable to Manage Uncertainty

- Robots freeze the moment reality drifts.
- Did it work the second time?

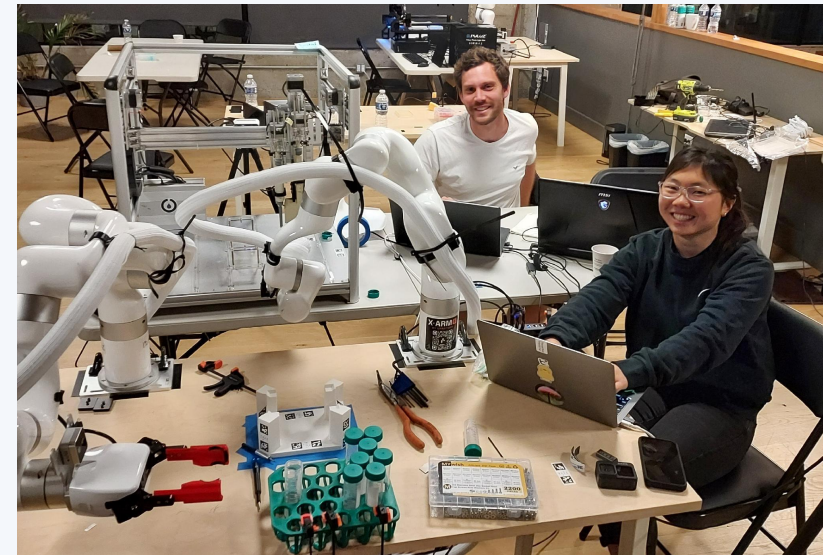
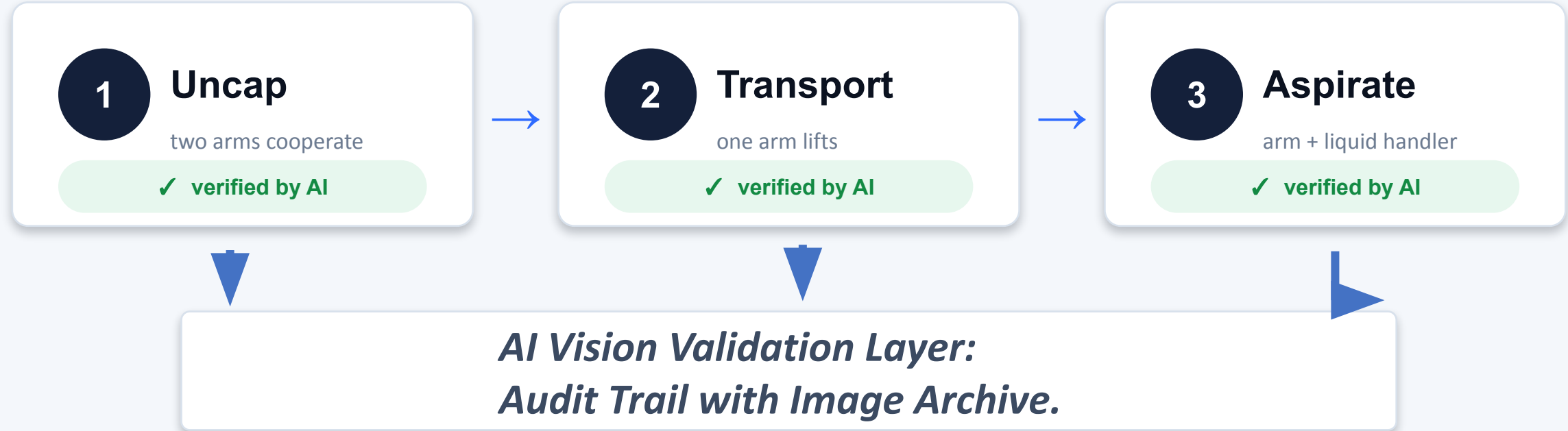
2

Tailored to Humans

- Instruments and labware are built for human hands:
 - Caps to twist
 - Tubes to hold
 - Liquids to pour
 - Robots can't just use them

We close both — dexterity, and awareness of successful completion.

A real sample-prep step — verified at every move



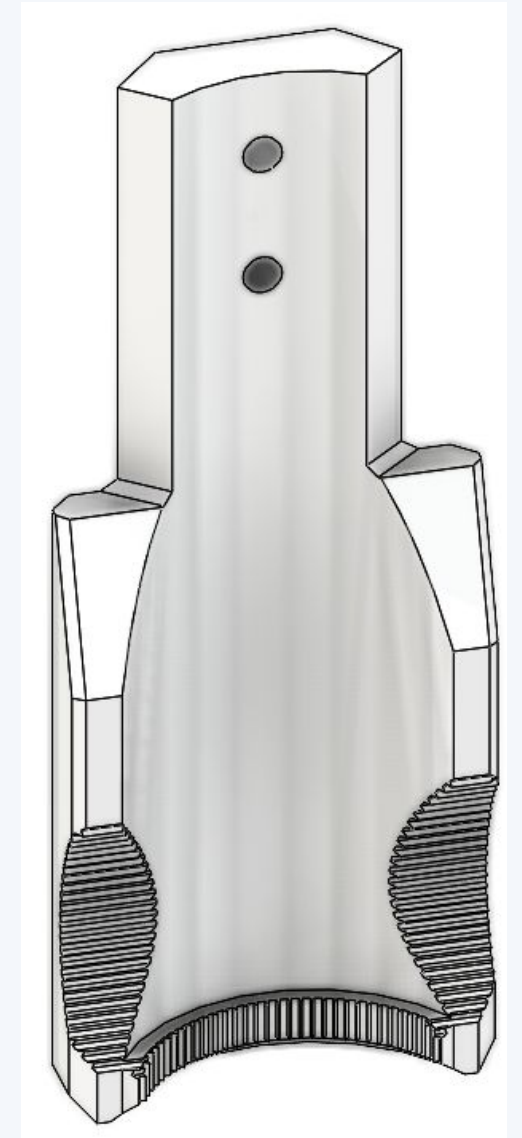
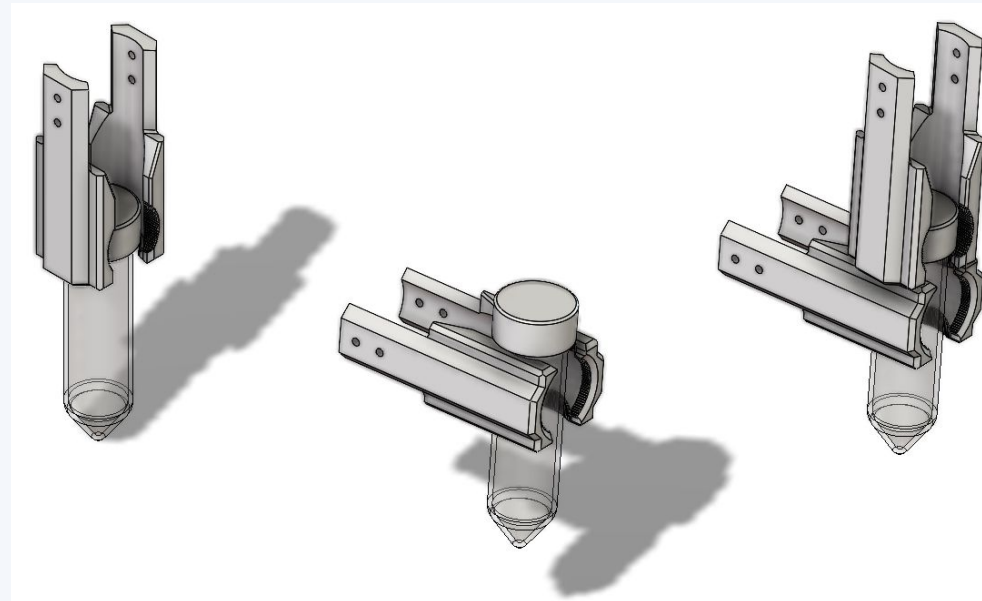
Custom Grippers

● Tight Space Grasping

Fit between other tubes.

● Two Orientation Grasping

A single design that grasps from the top or the side.



Cameras on every step



On-arm camera

Close-up view of the cap, grasp and tube opening — the fine detail verification needs.

External camera

Whole-scene view: where things are, and where they end up if something slips.

Feeds the agents

Every frame uses AI verification agent evaluation

READINESS

READY event stream live

all 15 spec waypoints taught

seq 119: resyncs 11 - gaps 0 - duplicates dropped 19 - a gap could not be repaired immediately — this view may be incomplete

SELECTED ACTION

follow the run again

0 initialize all devices 7/7 connected homed left right PLAN 54.2 s

aid 1 kind lifecycle: initialize speed slow real hardware started 22:48:31.211 finished 22:49:25.431

homed left, right no taught HOME none

device	connected	reality	detail
left	✓ connected	real	faults cleared, all axes enabled
right	✓ connected	real	faults cleared, all axes enabled
ot	✓ connected	real	driver initialize routine ran (axis wiggle, Z retracted)
overview_cam	✓ connected	real	captured a frame after 3 settle frames, stored 1280x720
handover_cam	✓ connected	real	captured a frame after 3 settle frames, stored 1920x1080
gripper_cam	✓ connected	real	captured a frame after 3 settle frames, stored 1280x720
gripper_left_cam	✓ connected	real	captured a frame after 3 settle frames, stored 1280x720

initialize · overview_cam · image/frame

initialize · handover_cam · image/frame

initialize · gripper_cam · image/frame

initialize · gripper_left_cam · image/frame

plan handover startup Start Pause (finishes current step) Resume Abort Inject

PLAN startup 1/1 done rev 1 follow the run collapse all (1)

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initialize · overview_cam · image/frame

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Rows are keyed on aid, not on the index: an injection rennumbers the indices after it, and a completed action's outputs must not vanish because its number changed.

Liquid Handler Integration



Responds to orchestration Tool

Coordinated actions between arm and liquid handler.

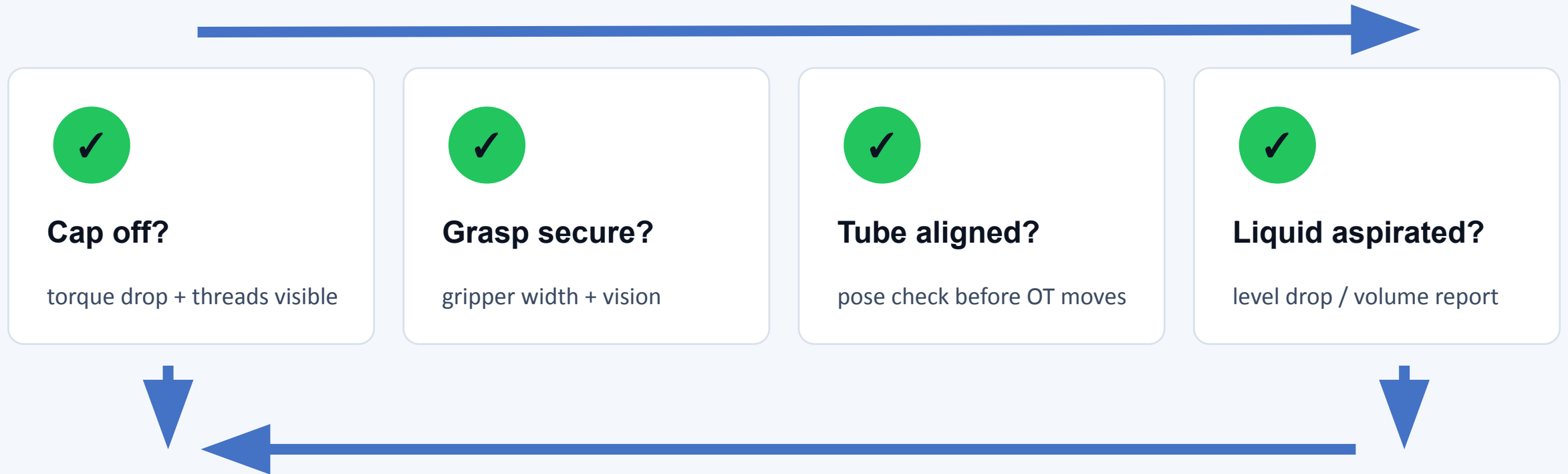


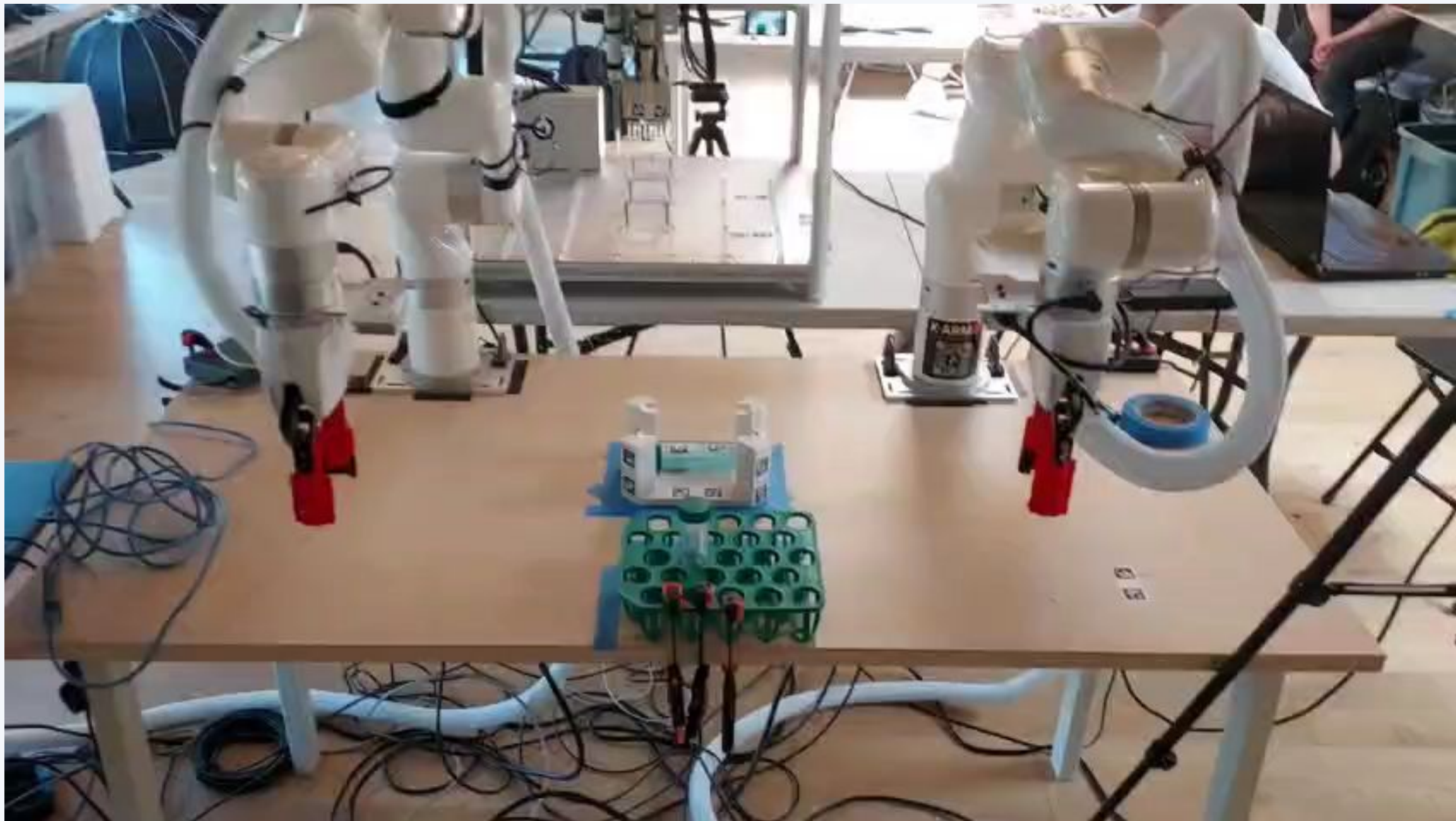
Opentrons 1 Liquid Handler...Enhanced!!

Integrates original liquid handling steps with camera verification.

AI Agents That Verify — And Retry

An agent confirms the action at each step, decides the recovery path when an error occurs, and learns for the next iteration.







Legacy Revival Integration with Original OT1

Back to the roots with first-generation equipment over 10 years old.



Thank you Chiu!

Chiu Chau

former CTO of Opentrons,
now CEO of OpenShelf — here at the hackathon.



BUILT IN 24H

The team



Lukas · Dale · Di

Lukas Bromig

Liquid handler & orchestration

co-founder, UniteLabs

Dale Herzog

Arm — dual xArm & end-effectors

CAD modeler · coffee nerd

Di Hu

Camera — perception & AI verify

bench scientist → AI×automation

github.com/Lbromig/hackathon-zeon